SQL Date Functions Exercise 25/06/2026

Q1)
```
SELECT
    order_id,
    customer_id,
    order_date,
    DAYNAME (order_date) AS day_name
FROM orders;
```

orders

Order_id	customer_id	order_date	day_name
1001	101	2026-05-01	Friday
1002	102	2026-05-02	Saturday
1003	103	2026-05-03	Sunday
1004	104	2026-05-04	Monday
1005	105	2026-05-05	Tuesday

2)
```
SELECT
    customer_id,
    customer_name,
    signup_date,
    MONTHNAME (signup_date) AS signup_month_name
FROM customer_signups;
```

customer_signups

customer_id	customer_name	Signup_date	sign_month_name
201	John	2026-01-15	January
202	Mary	2026-02-20	February
203	Peter	2026-03-05	March
204	Sarah	2026-04-18	April
205	Thabo	2026-05-30	May

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3) SELECT

```
SELECT
    sale_id,
    product_name,
    sale_date,
    MONTH(sale_date) AS sale_month
FROM sales;
```

sales

sale_id	product_name	sale_date	sale_month
S001	Laptop	2026-01-10	1
S002	Mouse	2026-02-15	2
S003	Keyboard	2026-03-20	3
S004	Monitor	2026-04-25	4
S005	Desk	2026-05-30	5

4) SELECT

```
SELECT
    transaction_id,
    customer_id,
    transaction_date,
    YEAR(transaction_date) AS transaction_year
FROM transactions;
```

transactions

transaction_id	customer_id	transaction_date	transaction_year
T001	101	2024-12-15	2024
T002	102	2025-01-20	2025
T003	103	2025-06-10	2025
T004	104	2026-02-05	2026
T005	105	2026-05-25	2026

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5)
```
SELECT delivery_id,
       customer_id,
       delivery_date,
       DAY (delivery_date) AS day_of_month
FROM   deliveries ;
```

deliveries

delivery_id	customer_id	delivery_date	day_of_month
D001	101	2026-05-01	1
D002	102	2026-05-08	8
D003	103	2026-05-15	15
D004	104	2026-05-22	22
D005	105	2026-05-29	29

6)
```
SELECT employee_id,
       employee_name,
       department,
       CURRENT_DATE ( ) AS today_date
FROM   employees;
```

employees

employee_id	employee_name	department	today_date
301	Nandi	Sales	2026-06-25
302	Brian	IT	2026-06-25
303	Lerato	Finance	2026-06-25
304	Sipho	HR	2026-06-25
305	Aisha	Marketing	2026-06-25

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7)
```
SELECT  order_id,
        customer_id,
        order_date_text,
        TO_DATE(order_date_text, 'YYYY-MM-DD') AS order_date
FROM    online_orders;
```

online_orders

order_id	customer_id	order_date_text	order_date
4001	101	2026-01-15	2026-01-15
4002	102	2026-02-20	2026-02-20
4003	103	2026-03-25	2026-03-25
4004	104	2026-04-10	2026-04-16
4005	105	2026-05-05	2026-05-05

8)
```
SELECT  payment_id,
        customer_id,
        payment_date,
        TO_CHAR(payment_date, 'YYYY-MM-DD') AS
        formatted_payment_date
FROM    payment_dates;
```

payment_dates

payment_id	customer_id	payment_date	formatted_payment_date
P001	101	2026-01-05	2026-01-05
P002	102	2026-02-10	2026-02-10
P003	103	2026-03-15	2026-03-15
P004	104	2026-04-20	2026-04-20
P005	105	2026-05-25	2026-05-25

9)
```
SELECT  customer_id,
        customer_name,
        last_purchase_date,
        DATEDIFF(current_date(), last_purchase_date) AS
```

days_since_last_purchase
FROM customer_purchases;

customer_purchases

customer_id	customer_name	last_purchase_date	days_since_last_purchase
501	John	2026-05-01	55
502	Mary	2026-05-10	46
503	Peter	2026-05-15	41
504	Sarah	2026-05-20	36
505	Thabo	2026-05-25	31

```
10) SELECT order_id,
           customer_id,
           order_date,
           DATEADD(order_date) AS expected_delivery_date
    FROM   shipping_orders;
```

shipping_orders

order_id	customer_id	order_date	expected_delivery_date
6001	101	2026-05-01	2026-05-08
6002	102	2026-05-03	2026-05-10
6003	103	2026-05-05	2026-05-12
6004	104	2026-05-07	2026-05-14
6005	105	2026-05-09	2026-05-16

```
11) SELECT booking_id,
           customer_id,
           booking_date,
           YEAR(booking_date) AS booking_year,
           MONTH(booking_date) AS booking_month,
           DAY(booking_date) AS booking_day
    FROM   bookings;
```

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bookings

booking_id	customer_id	booking_date	booking_year	booking_month	booking_day
B001	101	2026-01-12	2026	1	12
B002	102	2026-02-18	2026	2	18
B003	103	2026-03-22	2026	3	22
B004	104	2026-04-09	2026	4	9
B005	105	2026-05-27	2026	5	27

12)
```
SELECT order_id,
       customer_id,
       order_date,
       YEAR (order_date) AS order_year, amount
FROM   yearly_orders
WHERE  YEAR (order_date) = 2026;
```

yearly_orders

order_id	customer_id	order_date	order_year	amount
7004	104	2026-02-05	2026	1500
7005	105	2026-05-25	2026	2000

13)
```
SELECT order_id,
       customer_id,
       order_date,
       MONTH (order_date) AS order_month, amount
FROM   monthly_orders
WHERE  MONTH (order_date) = 3;
```

monthly_orders

order_id	customer_id	order_date	order_month	amount
8004	104	2026-03-25	3	1500
8005	105	2026-05-30	3	2000

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14)
```
SELECT subscription_id,
       customer_id,
       start_date,
       LAST_DAY (start_date) AS month_end_date
FROM   subscriptions;
```

subscriptions

subscription_id	customer_id	start_id	month-end
SUB 001	101	2026-01-10	2026-01-31
SUB 002	102	2026-02-15	2026-02-28
SUB 003	103	2026-03-20	2026-03-31
SUB 004	104	2026-04-25	2026-04-30
SUB 005	105	2026-05-30	2026-05-31

15)
```
SELECT send_id,
       customer_id,
       send_date,
       DATE_TRUNC('month', send_date) AS month_start_date
FROM   campaign_sends;
```

campaign_sends

send_id	customer_id	send_date	month_start_date
C001	101	2026-01-12	2026-01-01
C002	102	2026-02-18	2026-02-01
C003	103	2026-03-22	2026-03-01
C004	104	2026-04-09	2026-04-01
C005	105	2026-05-27	2026-05-01

16)
```
SELECT invoice_id,
       customer_id,
       invoice_date,
       TO_CHAR (invoice_date, 'Month YYYY') AS
       invoice_month_year
```

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```
FROM        invoice_dates;
```

invoice_dates

invoice_id	customer_id	invoice_date	invoice_month_year
INV001	101	2026-01-05	January 2026
INV002	102	2026-02-10	February 2026
INV003	103	2026-03-15	March 2026
INV004	104	2026-04-20	April 2026
INV005	105	2026-05-25	May 2026

17)
```
SELECT  customer_id,
        customer_name,
        date_of_birth,
        DATEDIFF (CURRENT_DATE (), date_of_birth) AS
        `customer_age
FROM    customer_birthdays;
```

customer_birthdays

customer_id	customer_name	date_of_birth	customer_age
901	John	1998-05-10	28
902	Mary	1990-08-20	35
903	Peter	2002-03-15	24
904	Sarah	1985-12-01	40
905	Thabo	2000-07-30	25

18)
```
SELECT  order_id,
        customer_id,
        order_date,
        DAYNAME (order_date) AS day_name,
      CASE
        WHEN DAYNAME (order_date) IN ('Saturday') THEN
                'Weekend'
        WHEN DAYNAME (order_date) IN ('Sunday') THEN
                'Weekend'
```

ELSE 'Weekday'
FROM weekend_orders;

weekend_orders

customer id	order_date	day_name	customer_id	day_type
9001	2026-05-01	Friday	101	Weekday
9002	2026-05-02	Saturday	102	Weekend
9003	2026-05-03	Sunday	103	Weekend
9004	2026-05-04	Monday	104	Weekday
9005	2026-05-05	Tuesday	105	Weekday

19) SELECT transaction_id,
customer_id,
transaction_date,
QUARTER (transaction_date) AS transaction_quarter, amount
FROM quarterly_transactions;

quarterly_transactions

transaction_id	customer_id	transaction_date	transaction_quarter	amount
Q001	101	2026-01-15	1	500
Q002	102	2026-03-20	1	1200
Q003	103	2026-04-10	2	800
Q004	104	2026-07-05	3	1500
Q005	105	2026-10-25	4	2000

20 SELECT order_id,
customer_id,
order_date,
DAEDIFF (CURRENT_DATE (), order_date) AS days_since_order, amount
FROM recent_orders
WHERE DATEDIFF (CURRENT_DATE (), order_date) > 30;

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recent_orders

order_id	customer_id	order_date	days_since_order	amount
R001	101	2026-04-01	85	500
R002	102	2026-04-15	71	1200
R003	103	2026-05-01	~~65~~ 55	800
R004	104	2026-05-10	~~56~~ 46	1500
R005	105	2026-05-25	31	2000

Bonus challenge

```
SELECT  customer_id,
        customer_name,
        last_purchase_date,
   CASE
   WHEN DATEDIFF (CURRENT_DATE(), last_purchase_date) ≤ 30
        THEN 'Active Customer'
   WHEN DATEDIFF (CURRENT_DAE (), last_purchase_date) BETWEEN
          31 AND 90 THEN 'At Risk Customer'
    ELSE 'Inactive Customer'
   END AS customer_status,
FROM    customer_recency;
```

customer_receny

customer_id	customer_name	last_purchase_date	days_since_last_purchase	customer_status
1001	John	2026-05-25	31	At Risk customer
1002	Mary	2026-05-10	46	At Risk customer
1003	Peter	2026-04-01	85	At Risk Customer
1004	Sarah	2026-02-15	130	Inactive customer
1005	Thabo	2025-12-20	187	Inactive customer

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